

Zain Khan

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Computer Engineering

Sep. 2025 – Apr. 2030

– Awarded over **\$40,000** in merit-based scholarships, including the **University of Waterloo Presidential Scholarship**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, C#, Java, Ruby, SQL, HTML/CSS

Frameworks: React, Redux, Next.js, Node.js, Express, Django, FastAPI, .NET/WPF, ROS 2, TensorFlow, OpenCV

Infrastructure: PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Linux, Git, Vercel, Playwright

EXPERIENCE

Leap Tools

Toronto, ON

Software Development Intern, Bespoke Team

May 2026 – Aug. 2026

- Built and shipped Surface Designer end-to-end (TypeScript, React/Redux, Next.js) on the Roomvo visualizer used in **80+ countries**: moved tile-pattern generation from a server-side C++ render service into the browser via a parametric format and one generic lattice-expansion algorithm, enabling live in-browser pattern editing with zero per-render round-trips; shipped **19** production patterns, now adopted by several companies.
- Delivered commerce features across **6** more production apps, integrating Python/Django microservices over HTTP and gRPC on PostgreSQL and AWS: CORS-enabled REST APIs serving visitor data to external vendor storefronts, **10**-event PostHog analytics, and a full-stack persistence slice with a Postgres/Kysely migration.
- Authored **402** tests including the app's first Playwright E2E suite (**21** specs); fixed an SSR-hydration flake class breaking E2E runs; moved the team onto agent harnesses with pre-commit gates.

WATonomous

Waterloo, ON

Robotics Software Engineer, EVE Autonomy

Jan. 2026 – Apr. 2026

- Developed concurrent autonomous robotics software in C++ on ROS 2 Humble, coordinating **5+ real-time nodes** across perception, planning, and control subsystems.
- Implemented costmaps, A* path planning, and Pure Pursuit control, achieving stable point-to-point navigation in cluttered indoor environments with dynamic obstacle avoidance.

Kasperia

Remote

Software Engineer (Contract)

Jun. 2023 – Apr. 2026

- Built **KPM (Kasperia Project Manager)**, a C#/.NET WPF desktop app that became the company's internal procurement operations system, running projects end-to-end from bid to supplier quotes, pricing, and order, with AI agents that draft supplier follow-ups and payment emails from live project state.
- Designed KPM's current-state + history data model, separating live fields (price, lead time, PO, payment) from per-aspect histories (quotes, payments, deliveries) so every value traces to its source.
- Rebuilt the client-facing website with reusable React components, improving page load by **30%** and cutting maintenance overhead by **25%**; built SQL/REST data pipelines that cut manual reporting time by **100%**.

PROJECTS

otto  | *Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase*

- Shipped an AI voice agent that turns natural speech into automated actions across GitHub, Gmail, and Google Calendar; awarded **1st Place — LiveKit Track** at NexHacks.
- Engineered a LiveKit WebRTC pipeline with Deepgram VAD turn detection and Gemini 2.5 Flash Realtime native audio inference, achieving **sub-second** voice-to-action latency; cut per-request tokens **4000 → 1800 (55%)**.

toothfairy  | *Next.js, TypeScript, FastAPI, Python, TensorFlow, OpenCV, Gemini 2.5 Flash, Moorcheh AI*

- Built a dental X-ray segmentation pipeline using a U-Net (TensorFlow/Keras) with OpenCV post-processing, isolating **28+ individual teeth** from panoramic radiographs.
- Orchestrated **3 AI copilots** (Gemini 2.5 Flash, Gemini Vision) for pathology detection across **10 dental conditions**, with per-patient RAG memory via Moorcheh AI.

extgen  | *React, TypeScript, Chrome Extensions, FastAPI, Python, Agentverse, uAgents, Gemini*

- Built a Chrome side panel that turns natural-language requests into installable Manifest V3 extensions, with an Edit DOM mode that exports live page edits into a permanent extension.
- Designed an Agentverse/uAgents pipeline with Architect, RAG, Codegen, Validator, and Packager agents, combining deterministic intent templates with LLM generation.